



NOVA DRIVE

TEAM REDFLAGS



SWOT ANALYSIS

Strength

- Environment friendly due to less GHG emission
- Low O&M cost
- Reduce the dependence on foreign oil imports
- Easy to drive
- Energy efficient transport medium

Weakness

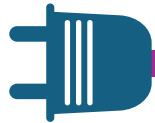
- Limited range and low speed vehicle
- Low O&M cost
- High charging time
- Battery need to change periodically
- Low consumer awareness
- Frequent road accident

Opportunities

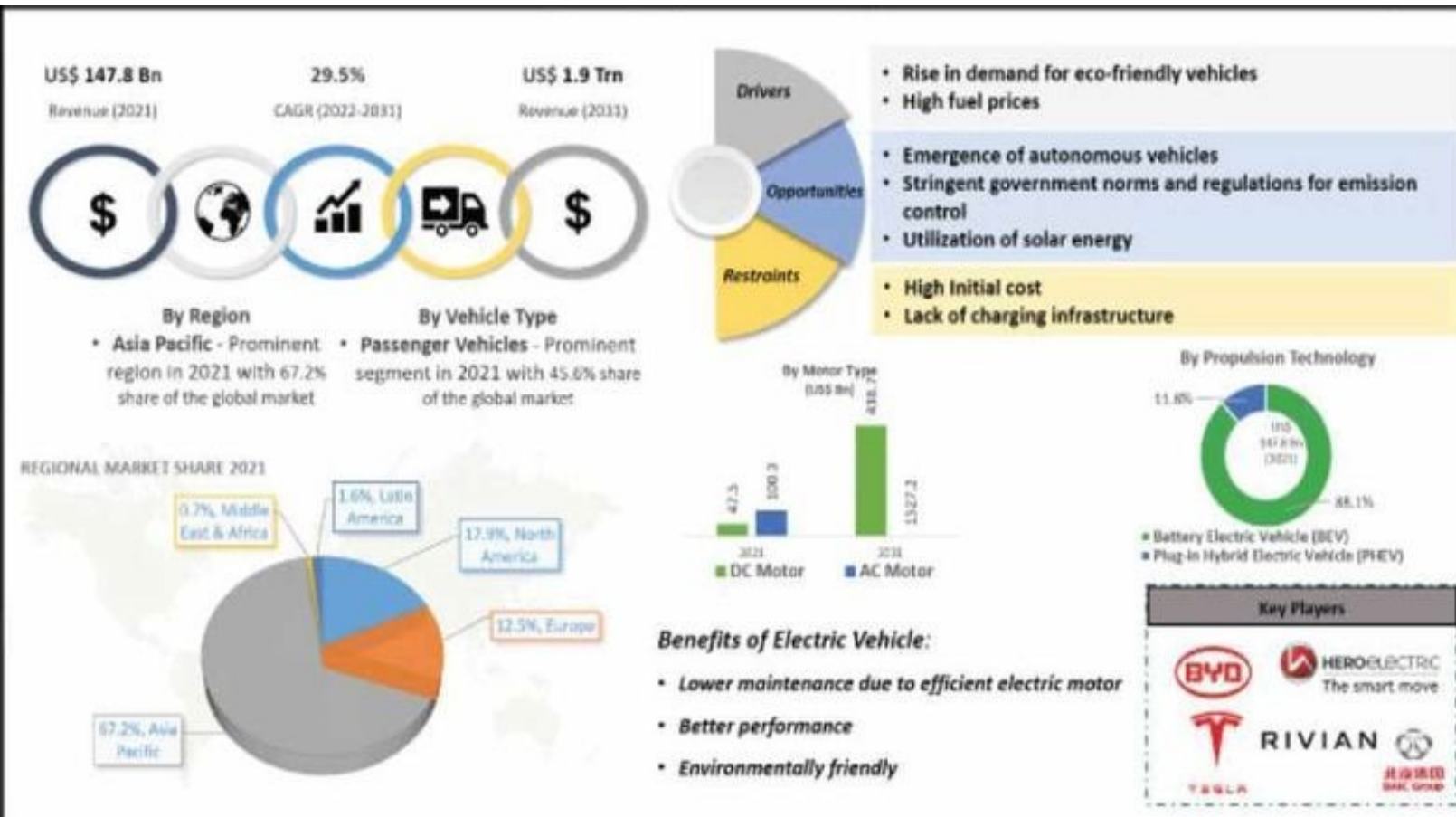
- Increase employment opportunities
- Availability of low cost labor
- Improvement in battery technology
- Integration of renewable resources for charging
- Rising fuel prices
- Rising awareness on environment sustainability

Threats

- Lack of Government support
- Available competitors in EV manufacturing
- Lack of charging station
- Insufficient spare parts
- Poor road and traffic condition



COMPETITOR MARKET ANALYSIS

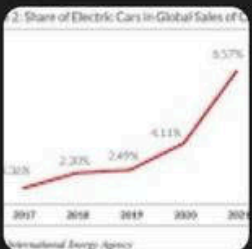


Full World

BD

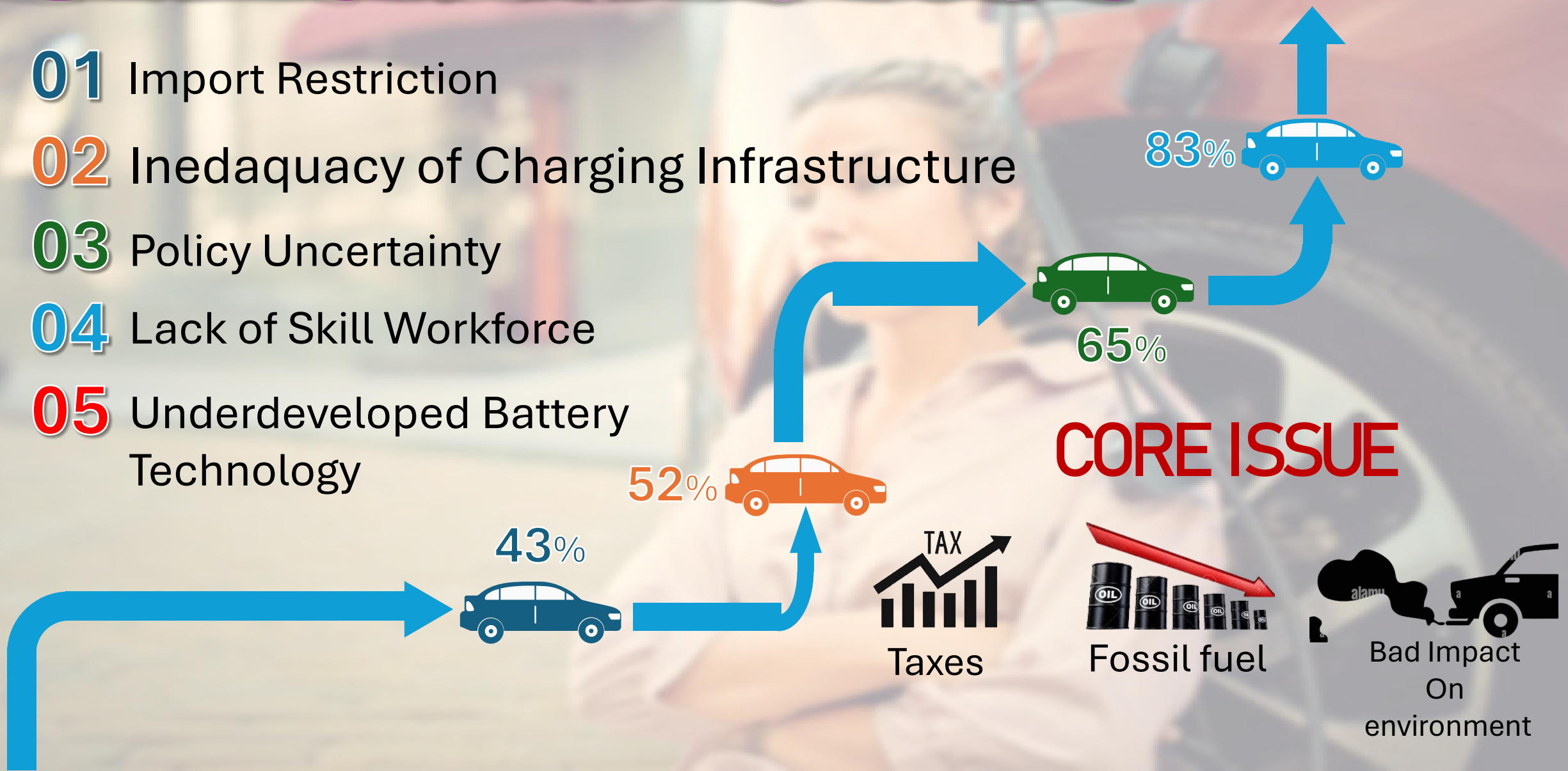
How big is the EV market in Bangladesh?

The all-electric vehicle market is almost nonexistent in Bangladesh. However, the country is experiencing consistent growth in the imports of hybrid electric vehicles. A growth of 154% in hybrid car imports from 3,296 units in FY18 to 8,366 units in FY21 has been observed.

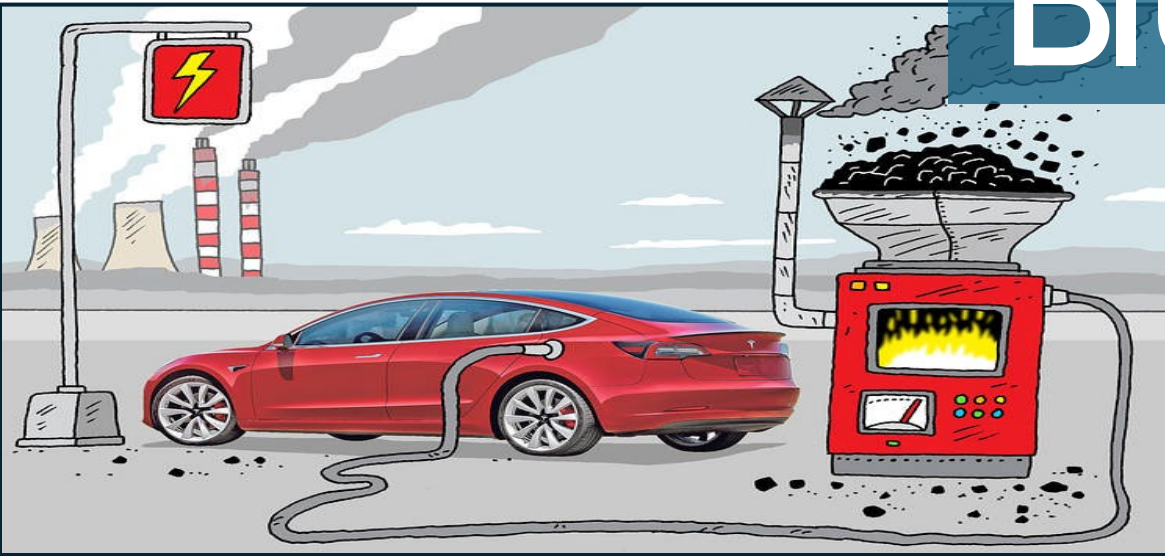


PROBLEM IDENTIFICATION

- 01 Import Restriction
- 02 Inadaquacy of Charging Infrastructure
- 03 Policy Uncertainty
- 04 Lack of Skill Workforce
- 05 Underdeveloped Battery Technology



BIG IDEA



Increasing rate of fuel cars will cause a lot of damage to the environment and also rise an issue about "non-renewable" energy sources



We will build a company of our own and produce electric vehicles since the tax rate is high for our country

REPLACEMENT



Seat system



ECU

Seat
adjustment
motor

Seat
air-conditioning
fan

Seat tactile
device

Door system



ECU

Door motor

Door-locking
door motor
module

Built-in
switch

Product strategy
in the age of CASE



Pump module



Electric oil pump
module

Electric water
pump module

Electric power steering system



EPS power pack
(column type)

EPS power pack
(rack type)

EV traction system

Lineup of Nidec's
traction motor system,
"E-Axle"



Ni150Ex
(150kW/3900Nm)
Mass production to start
in the second half of 2020 (plan)

Ni100Ex
(100kW/2400Nm)
Mass production to start
in 2022 (plan)

Ni70Ex
(70kW/1600Nm)
Mass production to start
in 2021 (plan)

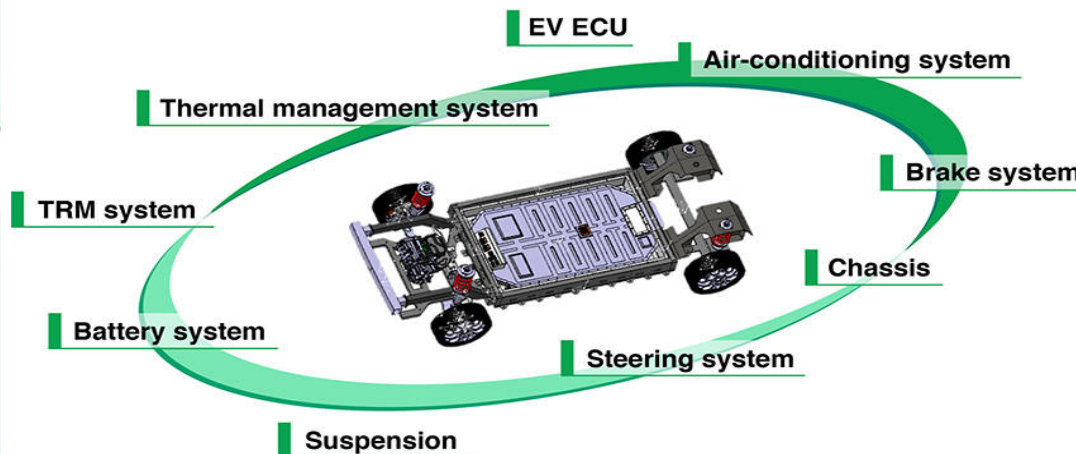
Ni50Ex
(50kW/1600Nm)
Mass production to start
in 2022 (plan)

Ni200Ex
(200kW/4200Nm)
Mass production to start
in 2023 (plan)

PRODUCT FEATURE

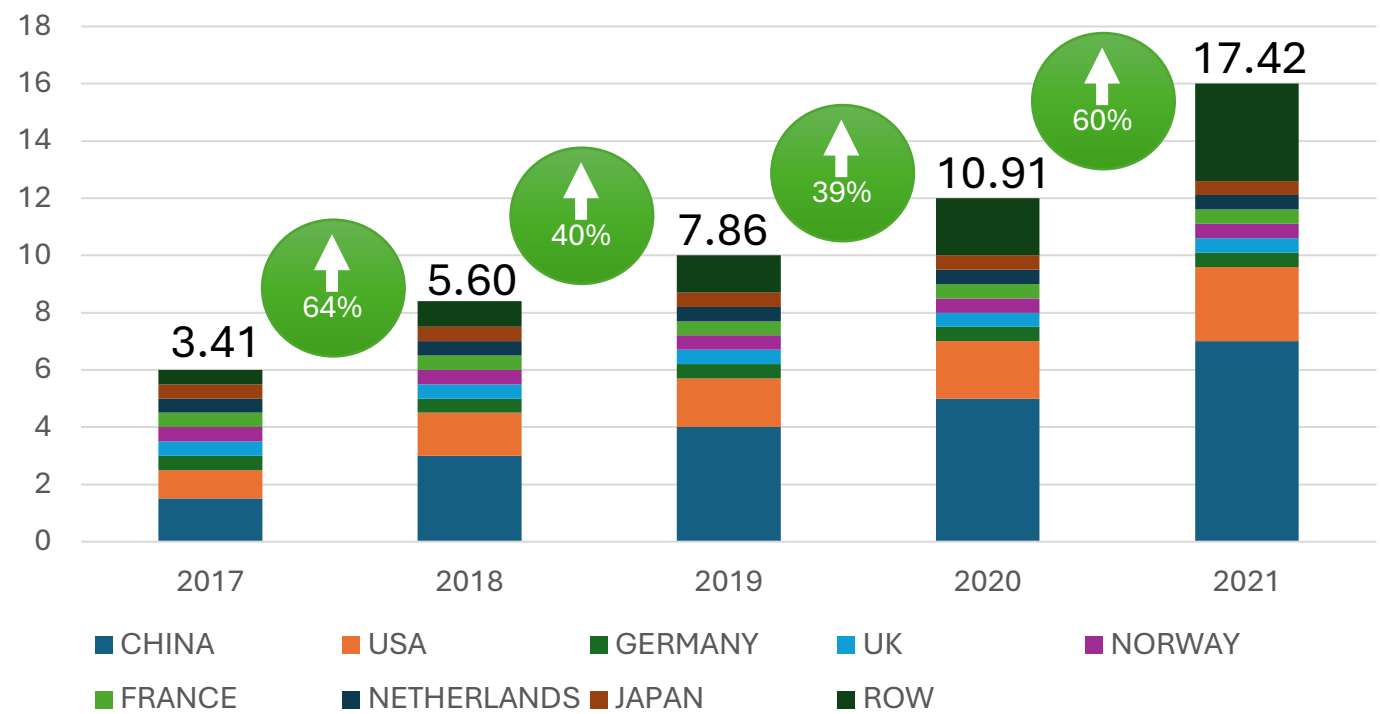


Best electric vehicle in the world



TARGET MARKET IDENTIFICATION

Stock of Electric Cars



BUSINESS CANVAS MODEL

VALUE PROPOSITIONS

- Best in Class-fully electric vehicles(Evs)
- Advance Electric vehicle Swap Batteries
- Little Business Competition

Customer Relationships

- Customer Service
- Brand
- Reputation

Key Resources

- Battery System
- Inverter
- Automation & Engineering

Customer Segments

- Commercial Vehicles-targeted towards transportation, shipping, and airplane business
- Green community
- High-income Society

Key Activities

- Design
- Software Development
- Maintenance of Charging Stations

Channel

- Swap Battery Area
- Self Service online store
- Retail And Service

KEY PARTNERSHIPS

- Government
- Dana Holding
- Major Suppliers
- Transportation company

Revenue Streams

- Automation Segment Revenue
- Energy Generation & Storage Segment
- Swap Batteries Application

Cost Structure

- Interest Expense, Taxes & Others
- Material, Labor, & Manufacturing Cost
- Marketing And Administration Cost
- Restructuring, Research & Development Cost

ELECTRIC VEHICLE COMPARISON

BENEFIT ANALYSIS

Electric Vehicle Charging Infrastructure



Level 1 and Level 2
Residential Charging



Level 2
Work and Public place Charging

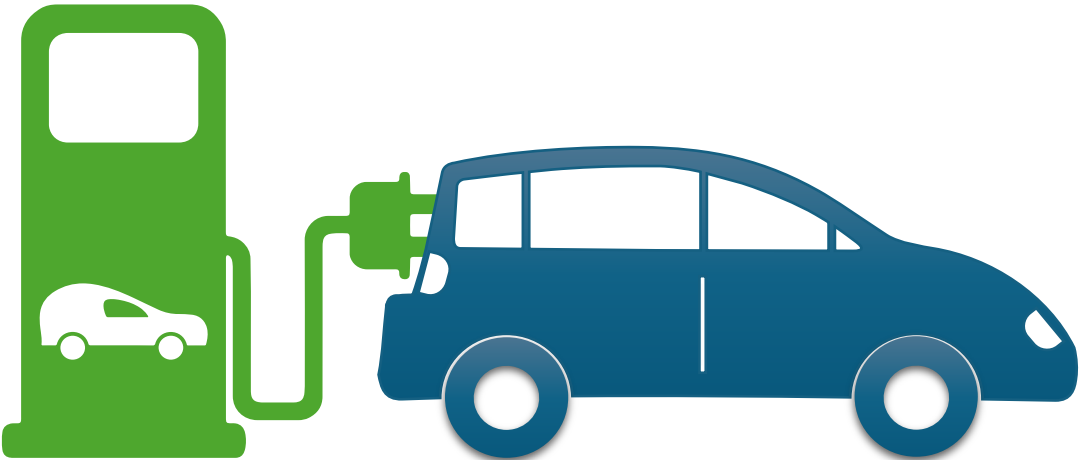


Level 3
DC Fast Charging

Electric vehicles are charged via an AC power supply at a normal (Level1) or semi fast charging rate:
Voltage
120V 1-Phase AC
Amps
12-16 Amps
Charging Loads
1.4 to 1.9 KW
Charging Time
3-5 Miles of range per hour
Price per Mile
2c-6c mile

Electric vehicles are charged via an AC power supply at semi fast (Level2) charging rate:
Voltage
208V or 240V 1-Phase AC
Amps
12-80 Amps (Typ 32 Amps)
Charging Loads
2.5 to 19.2KW (Type 7KW)
Charging Time
10-20 Miles of range per hour
Price per Mile
2c-6c mile

Electric vehicles are charged via an DC power supply at a fast (Level3) charging rate:
Voltage
208V or 480V 3-Phase AC
Amps
<125 Amps (Typ 60 Amps)
Charging Loads
<90KW (Type 50KW)
Charging Time
80% Charge in 20-32 minutes
Price per Mile
12c-25c per mile



Comparison of Electric Vehicle Types Considering Tank-to-Wheel Emissions and Energy-Ecological Efficiency

Materials and Methods

-  One vehicle for PHEV, HEV and PEV categories
-  Simulation of fossil fuels and biofuels combustion
-  Tank-to-Wheel (TTW) emissions evaluated considering the Brazilian electricity mix
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Findings



Highest values of TTW emissions were found for the PHEV



Considering the infrastructure and the use of biofuels, the HEV in dual-fuel mode is the most appropriate choice



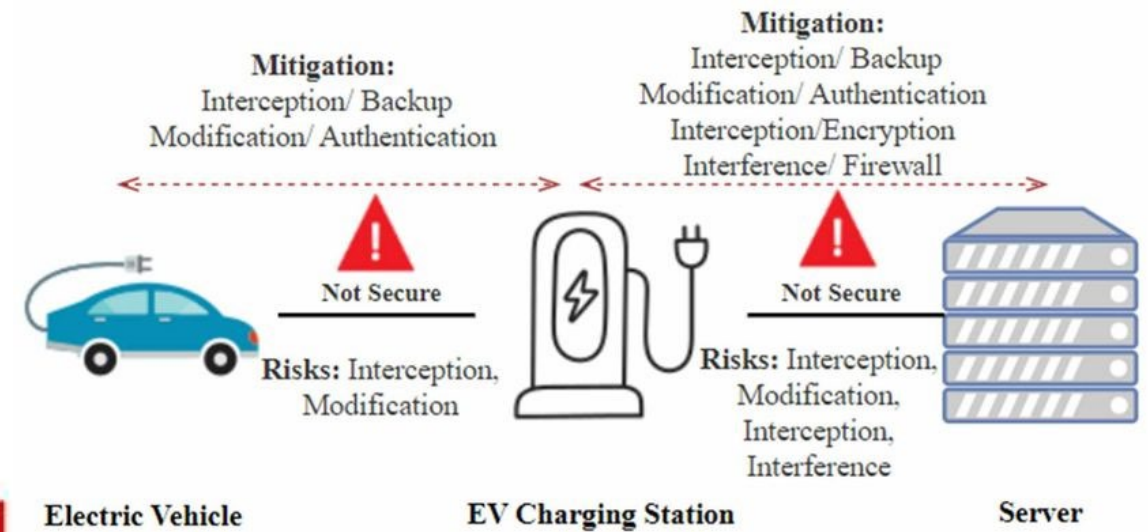
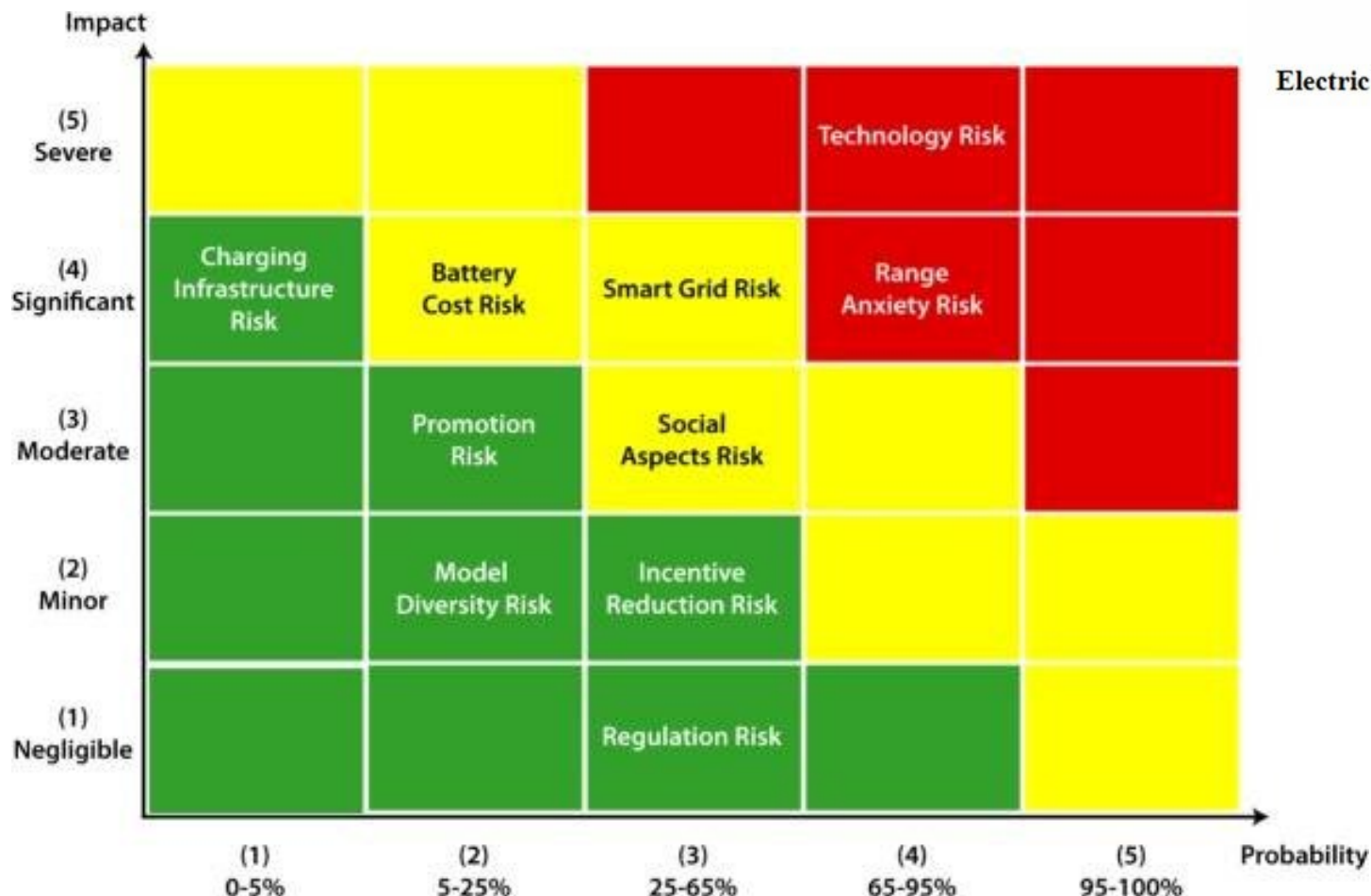
The PEV was the best choice to the Brazilian scenario when the emissions were considered



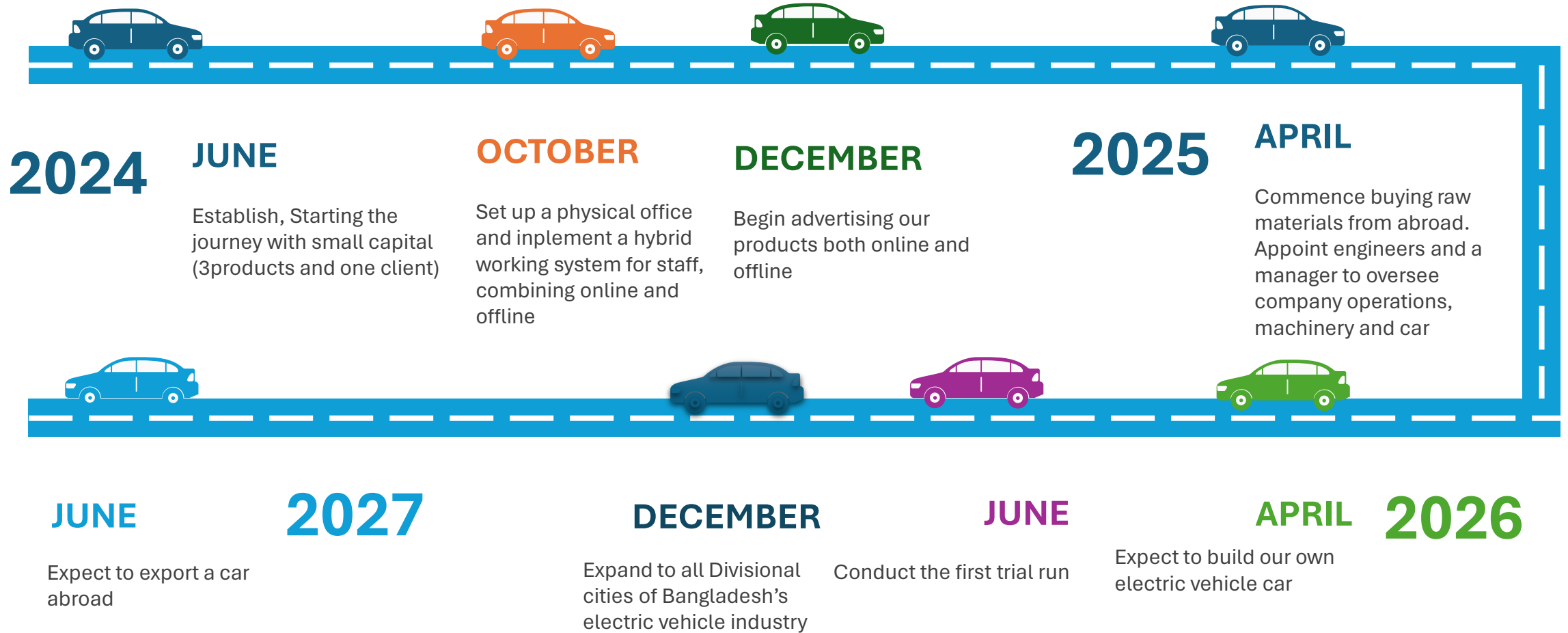
Highest value of energy-ecological efficiency was found for the engine in dual-fuel mode

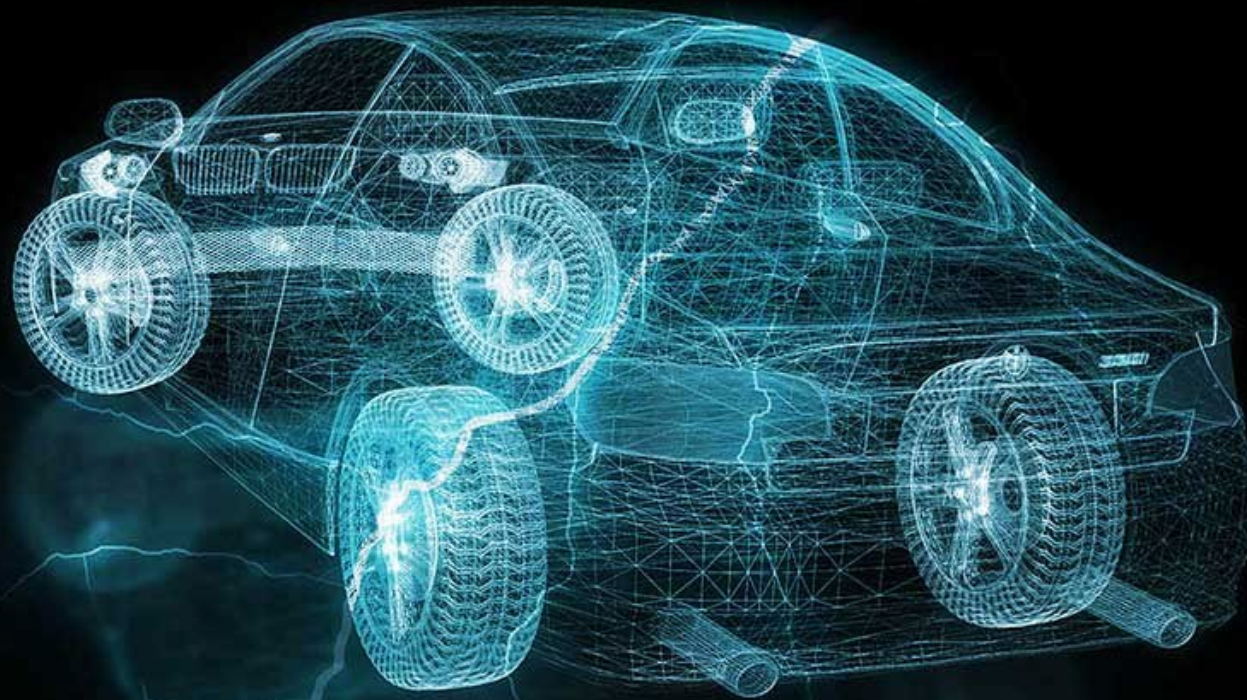
Conclusion: The findings presented address topics that will help in policies to achieve zero carbon dioxide emissions in transport sector. But, the dissemination of the electric vehicles depends on many factors, including government interest and investments.

RISK AND MITIGATION ANALYSIS



TIMELINE





THANK YOU